Battery-electric tractors for Värnamo, 2027: order 6

Recommendation. Order 6 battery-electric tractors under quotation Q-26-1187 for duties V02, V05, V09, V14, V16, V18. On the January design day each needs no more than 450 kWh against the 459 kWh FES-2 allows (85 per cent of 540 kWh usable). The heavy steel, ballast and metal duties V03, V07, V11 need about 470 kWh and stay on diesel, as do V21, V24, V27, V30, V33.

Duty	Shipper	Delivery	Round trip km	Planning load t	Design-day kWh	Electric
V02	Kulltorps Cykel AB	Kalmar	338	3.9	430	Yes
V05	Bredaryds Plast AB	Ängelholm	330	7.6	440	Yes
V09	Vaggeryds Glas AB	Göteborg	290	21.1	420	Yes
V14	Värnamo Isolering AB	Karlskrona	353	4.6	450	Yes
V16	Rydaholms Pall AB	Varberg	320	9.1	430	Yes
V18	Anderstorps Elektronik AB	Alingsås	337	5.4	440	Yes
V03	Gnosjö Stålcenter AB	Helsingborg	308	29.1	470	No
V07	Skillingaryds Ballast AB	Linköping	297	31.9	470	No
V11	Hillerstorps Metall AB	Kristianstad	312	26.5	470	No
V21	Reftele Trä AB	Malmö	345	18.3	490	No
V24	Vaggeryds Glas AB	Oskarshamn	327	21.0	480	No
V27	Gränna Dryckes AB	Trollhättan	357	13.5	490	No
V30	Bankeryds Mejeri ek. för.	Uddevalla	345	13.4	480	No
V33	Kulltorps Cykel AB	Västervik	384	3.9	490	No

Energies to the nearest 10 kWh, at −5 C and each duty's planning load (the mean of the last weighing of each CMR at Värnamo, November 2025 to October 2026, gross less tare). Limit 459 kWh under FES-2 4.3.

How the energy was estimated. From the 2334 duty-days the six Jönköping tractors ran in the pilot, joined to the TMS consignments, with distance, climb and temperature (below 16 C and a further rate below 0 C) as the other terms. The pilot's payloads are the shippers' declared weights. The Värnamo weighbridge shows the declarations are right on average but that the furniture, packaging, drinks and builders' merchants' clerks estimate them (their declared weight scatters around what the truck carries), the dairy writes one standard weight on every note, and Tabergs Trävaror prints its own weighbridge figure. Fitted on the declared weights, the model puts a tonne of payload at about 1.0 kWh per 100 km loaded, because the scatter in the declarations dilutes the effect. Taking each declaration to the weighed load it stands for, shipper by shipper, puts it at 2.2 kWh, which the Tabergs days confirm on their own.

What this changes. On the declared weights the heavy duties look feasible at about 440 kWh and the light insulation duty V14 does not (about 470 kWh): that would order 8 tractors and send 3 duties out that cannot finish a cold day on 85 per cent of the battery. The duty-by-duty figures are in ev_duty_assessment_2027.xlsx with the model and the calibration.

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